

Consulting Proposals - Phase 2

1. Key challenges and unmet needs.

Our client - Distribution System Operator - is asking for optimization-based support for planning network expansion amid rapid electrification. The main concern is optimal investment timing and prioritization of grid reinforcements. Key challenges that constraint the problem are:

- Uncertain, scenario-based demand growth
- Limited financial flexibility
- Reinforcements that are long-lived, meaning early decisions constrain later options

The objective is to minimize the total discounted investment cost over a 10-year horizon, while maintaining grid reliability.

Without optimization tools it is difficult for our client to understand and predict how uncertain demand growth scenarios and long-lived investment decisions affect optimal grid reinforcement planning for minimal investment costs and load reliability.

2. Proposed modeling roadmap, assumptions and insights

To address the client's concerns and deliver robust, optimization-based decision support, we propose a roadmap of three progressively detailed models that provide targeted and practical solutions, like the client requested.

Model 1 - Deterministic Baseline.

The first model is a single-period mixed-integer linear program (MILP) assuming perfect foresight and a fixed demand forecast. The decision variables define the timing and capacity of network reinforcements for each feeder or substation. The objective is to minimize the total discounted capital and operational costs subject to nodal capacity and annual budget constraints. Key inputs include reinforcement costs, network topology, and projected load growth. Main assumptions are perfect information, linear costs, and fixed demand without uncertainty. This model provides a transparent benchmark for regulators and identifies cost-optimal upgrades under ideal conditions.

Model 2 - Intertemporal and Anticipative Extension.

Building on the deterministic framework, the second model introduces multiple time periods to link investment decisions across years, allowing staged or postponed upgrades. Decision variables include the timing, scale, and sequencing of reinforcements as well as the carry-over of existing capacity. The model minimizes discounted system costs while satisfying intertemporal budget constraints and capacity limits. Inputs extend to multi-year demand projections and dynamic capital budgets, while assumptions include fixed reinforcement costs and deterministic demand evolution. This anticipative setup quantifies the benefits of delaying or phasing investments and reveals which assets are most sensitive to timing decisions and long-term growth uncertainty.

Model 3 - Uncertainty-Aware and Robust Optimization.

The third model extends the anticipative formulation to explicitly incorporate uncertainty through a two-stage stochastic or robust optimization framework. Decision variables include first-stage investment choices and second-stage recourse actions such as emergency reinforcements or controlled curtailments. Inputs consist of probabilistic or scenario-based demand forecasts, scenario probabilities, and reliability targets (e.g., 99%). The model assumes linearized power flow, fixed unit costs, and a representative set of demand scenarios capturing future variability. By enforcing probabilistic reliability constraints across all scenarios, the model quantifies trade-offs between cost and reliability, identifies "no-regret" reinforcements that appear in all futures, and highlights where flexible or modular investments add the most strategic value.